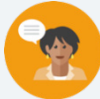


5.3 Exploring Angle-Angle-Side and Hypotenuse-Leg Triangle Congruence

Lesson Introduction

 Benchmarks	MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg. MA.912.GR.5.1 Construct a copy of a segment or angle.		 Learning Targets	- I can use constructions to determine if AAS congruence can be used to show that two triangles are congruent. - I can determine if HL congruence can be used to show that two triangles are congruent using deduction.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	- How can I use constructions to determine if AAS congruence can be used to show that two triangles are congruent? - How can I determine if HL congruence can be used to show that two triangles are congruent?			
 Lesson Narrative	In this lesson, students will continue their use of constructions to explore additional conditions for determining the congruence of two triangles. Students will use constructions to explore Angle-Angle-Side and Hypotenuse-Leg triangle congruence.			
 Component Summary	5.3.1 Warm-Up (~5 min.)	Reflection (<i>SE p. 234</i>)	5.3.4 Bring It Together (10 min.)	Formalized introduction to the Angle-Angle-Side (AAS) Congruence Theorem and the Hypotenuse-Leg (HL) Theorem (<i>SE p. 239</i>)
	5.3.2 Exploration (10-15 min.)	Exploration with different constructions to determine triangle congruency (<i>SE p. 235</i>)	5.3.5 Try It (5-10 min.)	Determining if two triangles are congruent based on given information and which postulate or theorem would be used (<i>SE p. 240</i>)
	5.3.3 Collaboration (10 min.)	Collaboration on reasoning arguments for the criteria for triangle congruence (<i>SE p. 237</i>)	5.3.6 Wrap-Up (~5 min.)	Self-reflection on the understanding of each congruence postulate or theorem (<i>SE p. 240</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Angle-Angle-Side (AAS) Congruence Theorem - Hypotenuse-Leg (HL) Theorem <u>Additional Terms:</u> N/A		Patty paper	
			Additional Preparation The design of the lesson is not to give students the information, but for them to discover the AAS Congruence Theorem and the Hypotenuse Leg Theorem. Consider doing the Exploration on your own to see how this unfolds to make it easier to connect with students.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not use the included angle when appropriate. - Students may forget to use the Hypotenuse Leg theorem as an option for proving congruence. - Students may misread side lengths using line segment notation.	- This lesson is primarily an exploration. Students will be using tools such as patty paper explicitly during the activity. Consider doing a brief review of what students did in the prior lessons as a refresher prior to beginning this lesson. - If time is an issue, consider assigning 5.3.5 Try It for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Clarify, Critique, Correct During 5.3.3 Collaboration, students analyze, reflect on, and improve the sample arguments on triangle congruence. When writing the improvements, students will clarify and add additional information to make the correct argument more convincing.	MA.K12.MTR.4.1: Discussion and Reflection In 5.3.4 Bring It Together, students will be working to bring together all they learned in 5.3.2 Exploration to make a conjecture and learn the formalized definitions of Angle-Angle-Side and Hypotenuse-Leg.
 Differentiate Instruction	Consider pairing students based on their performance in the prior lesson. The data from the Wrap-Up activity from Lesson 1 can serve as strong data for student groupings for this lesson. Consider the extended use of the patty paper when showing congruence throughout the lesson. Consider extended exploration time with students in small groups who need additional support. 5.3.5 Try It provides students with an opportunity to demonstrate mastery of determining if two triangles are congruent, including what theorem or postulate was used to make that determination. Consider using the student's labeling of the given information and the naming of the parts on the triangles as data when determining the level of mastery of congruent triangles.	
Lesson Notes:		

5.3.1 Warm-Up: Giving Back

Component Overview: Students explore a case study on working through challenges to succeed and giving back.



5.3.1 Warm-Up: Giving Back

Morgan State University is a historically black, public research university in Baltimore, Maryland. In the early 1960s, Calvin E. Tyler Jr., a young African



American man who was the first in his family to go to college, attended Morgan State. However, in 1963 he was forced to drop out when he was unable to pay his tuition.

He began working as a truck driver for the United Parcel Service (UPS). Throughout his career at UPS, he continued to be promoted and ultimately served as both the senior vice president for U.S. operations and as a director within the company. On February 23, 2021, he and his wife Tina pledged \$20 million to Morgan State University's endowment to go toward scholarships for students in financial need.



Choose one of the following questions to answer.

1. How do you think having a strong drive to succeed likely played a role in Mr. Tyler's career?

Answers will vary.

2. Why do you think it was important to Mr. Tyler to pledge this endowment?

Answers will vary.



- What obstacles have made math challenging for you so far?
- What connections are you making between Mr. Tyler's career and your journey in mathematics?

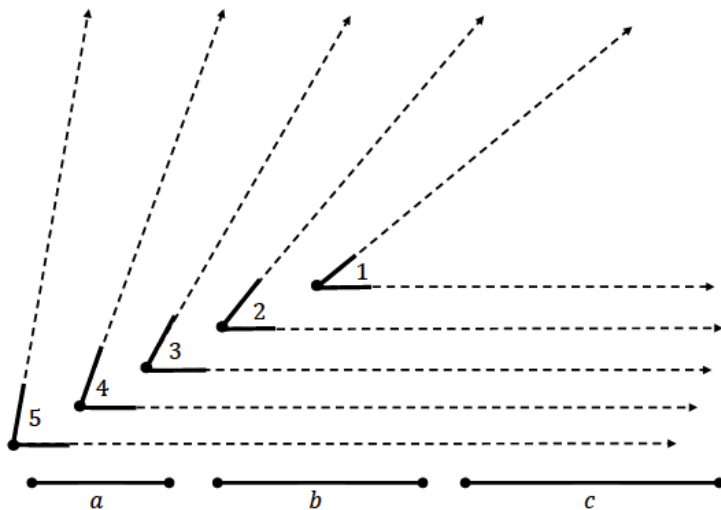
5.3.2 Exploration: Which Other Criteria Is Enough?

Component Overview: Students will explore different constructions to determine using new criteria for triangle congruency. For this activity, consider having each partner pair or small group work on one question at a time before bringing the class back together to discuss what they did, in order to ensure students are on the right track prior to progressing through the next part of the activity.



5.3.2 Exploration: Which Other Criteria Is Enough?

Five angles labeled 1-5 and three line segments labeled a , b , and c are given.



Use the given parts to create triangles by copying the parts needed onto patty paper using the given criteria below.

Helpful Hint: When copying an angle, include some (or all) of the dotted lines so that angles and line segments can be connected.



For this activity, consider providing students with a review of the instructions and setup for this problem. Do a verbal review of the angles and line segments provided on this page.



Notice and Wonder

- What do you notice about each of the angles provided? What about the line segments?
- What do you wonder about each of the angles provided? What about the line segments?

5.3.2 Exploration: Which Other Criteria Is Enough? (continued)

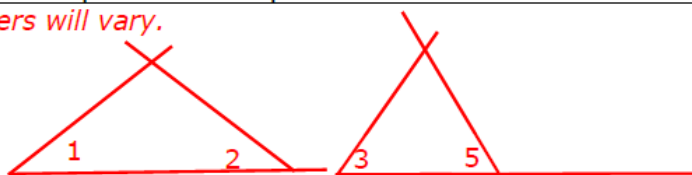
- Construct two different triangles using each set of criteria. Use the provided steps in each criteria.

Criteria 1:

Given two angles (AA)

- Step 1: Copy $\angle 1$ onto a piece of patty paper.
 Step 2: Close the triangle using $\angle 2$.
 Step 3: Trace the completed triangle. Label $\angle 1$ and $\angle 2$ in the completed triangle.
 Step 4: Label your triangle "Criteria 1 (AA)."
 Step 5: Repeat these steps with $\angle 3$ and $\angle 5$.

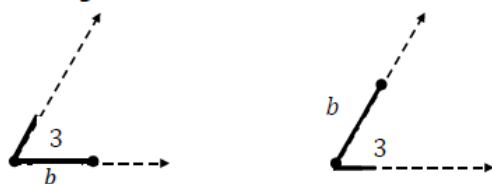
Answers will vary.



Criteria 2:

Given two angles and a non-included side (AAS)

- Step 1: Copy side b onto a piece of patty paper. Label side b on your patty paper.
 Step 2: Place $\angle 3$ at one of the ends of side b and trace the ray that does not overlap with side b to copy the angle. Label $\angle 3$.



- Step 3: Use $\angle 4$ to close the triangle such that $\angle 4$ is opposite from side b . Label $\angle 4$.
 Step 4: Trace the completed triangle. Label your triangle "Criteria 2 (AAS)."
 Step 5: Repeat these steps with side c , $\angle 1$, and $\angle 2$.



Avoid assisting students with locating the placement of $\angle 2$ on the side of $\angle 1$ (likewise for $\angle 5$ and $\angle 3$). Students should naturally have different placements of this second angle since the length of the side of the first angle is not specified.

Teachers may want students to tape their patty paper below each criterion.

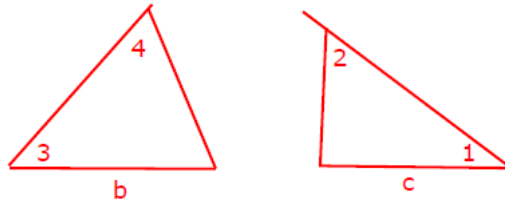


What are some things that you noticed about your triangles in Criteria 1?



Consider walking students through Criteria 2. The instructions may confuse students in the process. Guide students through each step, one-by-one, modeling each step and ensuring that students are completing each step correctly.

5.3.2 Exploration: Which Other Criteria Is Enough? (continued)



- Compare your four triangles with your partner's.
- Write your partner's name on the line in the table. Then, state "true" or "false" to indicate if your triangle is congruent to your partner's triangle for each criteria.

Criteria	AA		AAS	
	$\angle 1$ and $\angle 2$	$\angle 3$ and $\angle 5$	side b , $\angle 3$, and $\angle 4$	side c , $\angle 1$, and $\angle 2$
My triangle is congruent to <u>Partner</u> 's triangle.	Answers will vary. False	Answers will vary. False	True	True

- With your partner, find another pair of students to form a group of four. Write the two new students' names in the table below. Then, state "true" or "false" to indicate if your triangle is congruent to your partner's triangle for each criteria.

Criteria	AA		AAS	
	$\angle 1$ and $\angle 2$	$\angle 3$ and $\angle 5$	side b , $\angle 3$, and $\angle 4$	side c , $\angle 1$, and $\angle 2$
My triangle is congruent to <u>Student</u> 's triangle.	Answers will vary. False	Answers will vary. False	True	True
My triangle is congruent to <u>Student</u> 's triangle.	Answers will vary. False	Answers will vary. False	True	True



Students are not writing for question 2; they're just discussing the comparison. Their discussion will be summarized in the tables for questions 3, 4, and 5. The answer written here is guidance for what to listen for while monitoring the student discussion. The triangles in Criteria 1 are the same shape but different sizes. Highlighting this as a "notice" will build on the future learning of triangle similarity with AA.



Discussion questions for question 2:

- What is similar about all the triangles that follow Criteria 1?
- What did you notice about the triangles created under Criteria 2?
- What is the relationship between the triangles in Criteria 1 and Criteria 2?



Consider using physical movement in this process by having students move about the classroom in order to find an additional student pair.

5.3.2 Exploration: Which Other Criteria Is Enough? (continued)

5. Discuss with your group: If given either set of criteria, does that mean all of the parts of the triangles are congruent? Then, write your explanation in the table.

Criteria	Explanation
Given two angles (AA)	<i>Answers will vary. The triangles created with two congruent angles are not necessarily congruent. The triangles can be created different ways with different side lengths.</i>
Given two angles and non-included side (AAS)	<i>Answers will vary. If two adjacent angles in a triangle are congruent, then their third angle is also congruent because all triangles have a sum of 180°. If the triangles, then also have a corresponding side that is congruent, they will be congruent by ASA Congruence Theorem.</i>



Allow students to share with the entire class their explanation. Have students decide if they agree or disagree with the explanation, clarifying and refining their classmates' explanations when necessary.

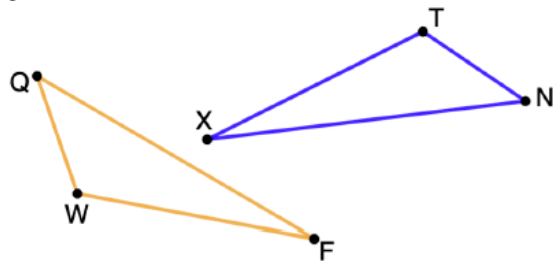
5.3.3 Collaboration: Making Connections

Component Overview: Students will work together to make connections with criteria for triangle congruency. For this activity, release students to work on the activity with a partner, without extensive prompting.



5.3.3 Collaboration: Making Connections

1. $\triangle QWF$ and $\triangle NTX$ are shown. $\overline{FQ} \cong \overline{XN}$, $\angle FQW \cong \angle XNT$, and $\angle QWF \cong \angle NTX$.



Two students discussed the information given.

Mabel thinks there is not enough information to determine congruence, because the side given is not between the two angles given.

Xavier thinks the triangles are congruent because he knows that the third angles in the triangles have to be congruent, and then there would be a side between two angles.

With your partner, discuss both students' reasoning. Then, answer the following.

- Explain whether you agree with Mabel or Xavier.
Answers will vary. Xavier is correct. Xavier understands that the side does not have to be included between the two angles to determine congruence.
- What additional information could be added to the argument you agree with to make it more convincing?
Answers will vary. Xavier could continue to explain that if two angles are congruent, the third must also be congruent. Knowing this he may point out that any triangle with 2 congruent angles and a congruent side will either fall into AAS or ASA. Both are able to prove congruence.



Consider allowing students to share where they believe the thinking for both Mabel and Xavier is coming from. Have students vote on whether they agree with Mabel or with Xavier.



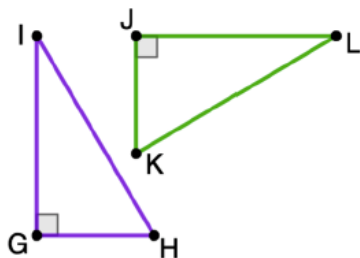
- Where does Mabel have an error in her thinking?
- If she asked you for help, what information would you tell her to help correct her?



- How do you know additional information is needed for Xavier's explanation?
- How would you know what specific additional information would be needed?
- How would that strengthen his case?

5.3.3 Collaboration: Making Connections (continued)

2. Triangle IHG and $\triangle LKJ$ are shown. $\overline{IH} \cong \overline{LK}$, $\overline{HG} \cong \overline{KJ}$, and $\angle IGH$ and $\angle LJK$ are right angles.



Eli and Kaylah had the following discussion.
Eli said, "In a situation where you have two sides without the angle in between them, you can't tell if the triangles are congruent."
Kaylah said, "Normally I would agree with you, except these are right triangles."

- a. Discuss with your partner what Kaylah might be thinking. Write a summary of your discussion.

Answers will vary. Kaylah realizes that because the triangles are right triangles the Pythagorean Theorem can be used to find the unknown third sides and thus show they are congruent.

- b. When given two sides of a right triangle, how can the length of the third side be determined?

By using the Pythagorean Theorem $a^2 + b^2 = c^2$



- How does Kaylah's opinion differ from that of Eli?
- Why would she think that the triangles being right triangles would cause a difference in Eli's definition?



Consider leveraging an Agree or Disagree instructional protocol. First, have students determine whether they agree or disagree with Kaylah's argument. Have students share out their rationale for why they chose to agree or disagree with Kaylah.



For an additional challenge, have students write a paragraph proof showing the two triangles are congruent from the given information.

5.3.3 Collaboration: Making Connections (continued)

Kaylah continued, "For a right triangle, as long as I have two of the sides, I can find the length of the third side. Then I can use SSS since all three corresponding sides would be congruent."

- c. Work with your partner to explain why Kaylah's argument could work to show congruence for any pair of right triangles where the hypotenuse and one leg are identified as congruent.

Answers will vary. Kaylah would be correct because the Pythagorean Theorem can be used to find the missing side. Since the other two sides of the triangle are congruent, the third side will also be congruent allowing us to connect with SSS congruence postulate.

- d. Explain whether Kaylah's argument would be needed to show congruence between right triangles where both legs of the triangles are identified as congruent.

Answers will vary. No Kaylah's argument would not be needed to show congruence between the right triangles. Given congruence of both legs of the triangle and the included right angle we can prove congruence under SAS.



If needed, consider probing student thinking as they collaborate to make connections:

- *How does Kaylah's reasoning about right triangles relate to SSS Congruence?*
- *How does Kaylah's reasoning about right triangles relate to SAS Congruence?*



Listen as students discuss questions 2c and 2d to determine misconceptions that may need to be addressed as well as student ideas to be shared with the whole group. Consider facilitating a whole group discussion to wrap up these ideas and address misconceptions prior to moving on to the next component.

5.3.4 Bring It Together: Triangle Congruence

Component Overview: Students will connect what they have learned through their exploration to the formal definition of the Angle-Angle-Side (AAS) Congruence Theorem and the Hypotenuse-Leg (HL) Theorem. For this activity, provide the narrative of the AAS definition and release students to work with their partners on questions 1 and 2. Provide the narrative of the Hypotenuse-Leg Theorem and release students to complete question 3.



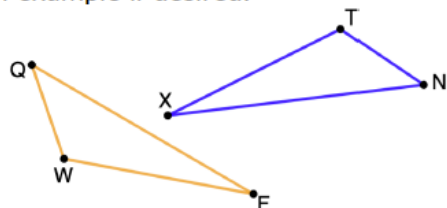
5.3.4 Bring It Together: Triangle Congruence



Angle-Angle-Side (AAS) Congruence Theorem

If two angles and a side opposite one of them are congruent to two angles and the corresponding side of the other, respectively, then the triangles are congruent.

1. Work with your partner to explain how Angle-Angle-Side Congruence Theorem is true. Use the triangles shown to provide an example if desired.



Answers will vary. AAS Congruence Theorem must be true because any two triangles that have 2 congruent angles must also have the third angles congruent. Since we also have any one side congruent, we could prove congruence also using the ASA Congruence Theorem.



When introducing the Angle-Angle-Side (AAS) Congruence Theorem, tie it back to what students have explored in the prior activity so that they can connect the formal definition to their discovery from earlier.



Have students share their thinking to question 1. This will provide all students with an opportunity to hear other explanations from their peers, but also provide you with potential misconceptions that can be addressed.

5.3.4 Bring It Together: Triangle Congruence (continued)

2. Complete the statements.

If two angles of one triangle are congruent to two angles of another triangle, then the third pair of angles are ☐ similar.
☒ congruent. If a congruent side opposite one of the given angles is congruent to the corresponding side of the other triangle, then ☐ SSS
☐ SAS
☒ ASA also applies. Thus,

AAS is enough to show congruence.



The Hypotenuse-Leg (HL) Theorem

Two right triangles are congruent if the hypotenuse and a leg of one are congruent to the hypotenuse and a leg of the other, respectively.

3. The previous lesson's exploration showed that SSA is not sufficient to prove congruence. Explain why HL is sufficient to prove congruence for right triangles, despite being two sides with a non-included angle.
Answers will vary. HL is sufficient for right triangles because I can find the length of the other leg using the Pythagorean Theorem, confirming that the third side of both triangles are congruent.



When introducing the Hypotenuse-Leg (HL) Theorem, tie it back to what students have explored in the prior activity so that they can connect the formal definition with their discovery from earlier.



Would HL work in situations where the triangles are not right triangles? Why or why not?

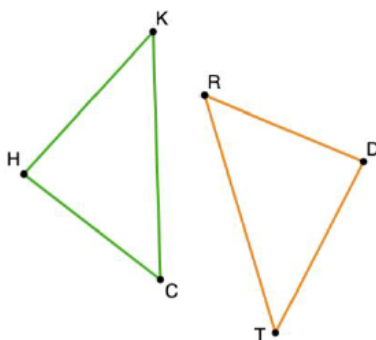
5.3.5 Try It: Which Criteria?

Component Overview: Students will practice determining if two triangles are congruent and which postulate or theorem would be used. This activity can be done independently to determine student understanding of triangle congruency criteria.



5.3.5 Try It: Which Criteria?

1. Triangles HKC and DTR are shown. Determine if the given information is sufficient to determine the triangles are congruent. If so, name the congruence theorem or postulate that shows that.



For this activity, begin by providing students with the direction. It may be necessary to remind them that this table includes all theorems and postulates they've learned thus far, including this lesson.

Given Information	Are They Congruent?	Theorem or Postulate Used
$\overline{HK} \cong \overline{DT}$, $\overline{KC} \cong \overline{TR}$, $\angle KCH \cong \angle TRD$	No	
$\angle KCH \cong \angle TRD$, $\angle HKC \cong \angle DTR$, $\angle CHK \cong \angle RDT$	No	
$\angle KCH \cong \angle TRD$, $\angle HKC \cong \angle DTR$, $\overline{HC} \cong \overline{DR}$	Yes	AAS
$\overline{HK} \cong \overline{DT}$, $\overline{KC} \cong \overline{TR}$, $\overline{HK} \perp \overline{HC}$, $\overline{DT} \perp \overline{DR}$	Yes	HL
$\overline{HK} \cong \overline{DT}$, $\overline{HC} \cong \overline{DR}$, $\angle KHC \cong \angle TDR$	Yes	SAS



To differentiate the product, consider using the students' labeling of the given information and the naming of the parts on the triangles when determining the level of mastery of congruent triangles.



For a challenge, have students create triangles and/or conditions that would result in triangles being congruent under a theorem or postulate not used in the table.

5.3.6 Wrap-Up: Reflection

Component Overview: Students will complete a self-reflection on their level of understanding for each congruence postulate or theorem. Students should complete this activity independently in order for the instructor to get an accurate reflection of student understanding.



5.3.6 Wrap-Up: Reflection

- Circle the icon that best describes your understanding of each congruence postulate or theorem.



I need help.



I got it.



I could help someone.

Side-Side-Side (SSS)			
Side-Angle-Side (SAS)			
Angle-Side-Angle (ASA)			
Angle-Angle-Side (AAS)			
Hypotenuse-Leg (HL)			

Answers will vary.



This activity provides students with an opportunity to complete a self-reflection on their level of understanding of each congruence postulate or theorem they have learned so far in the unit. This data, coupled with unit performance, can provide valuable insight into which students may need additional support.